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References Book:

1. **Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann Publishers, third edition ,2013**
2. **Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar, Introduction to Data Mining, second edition, Pearson, 2019**
3. **Ian. H. Witten, Eibe Frank and Mark. A. Hall, Data Mining: Practical Machine Learning Tools and Techniques, third edition , 2017**

FACULTY HOD DEAN ACADEMICS

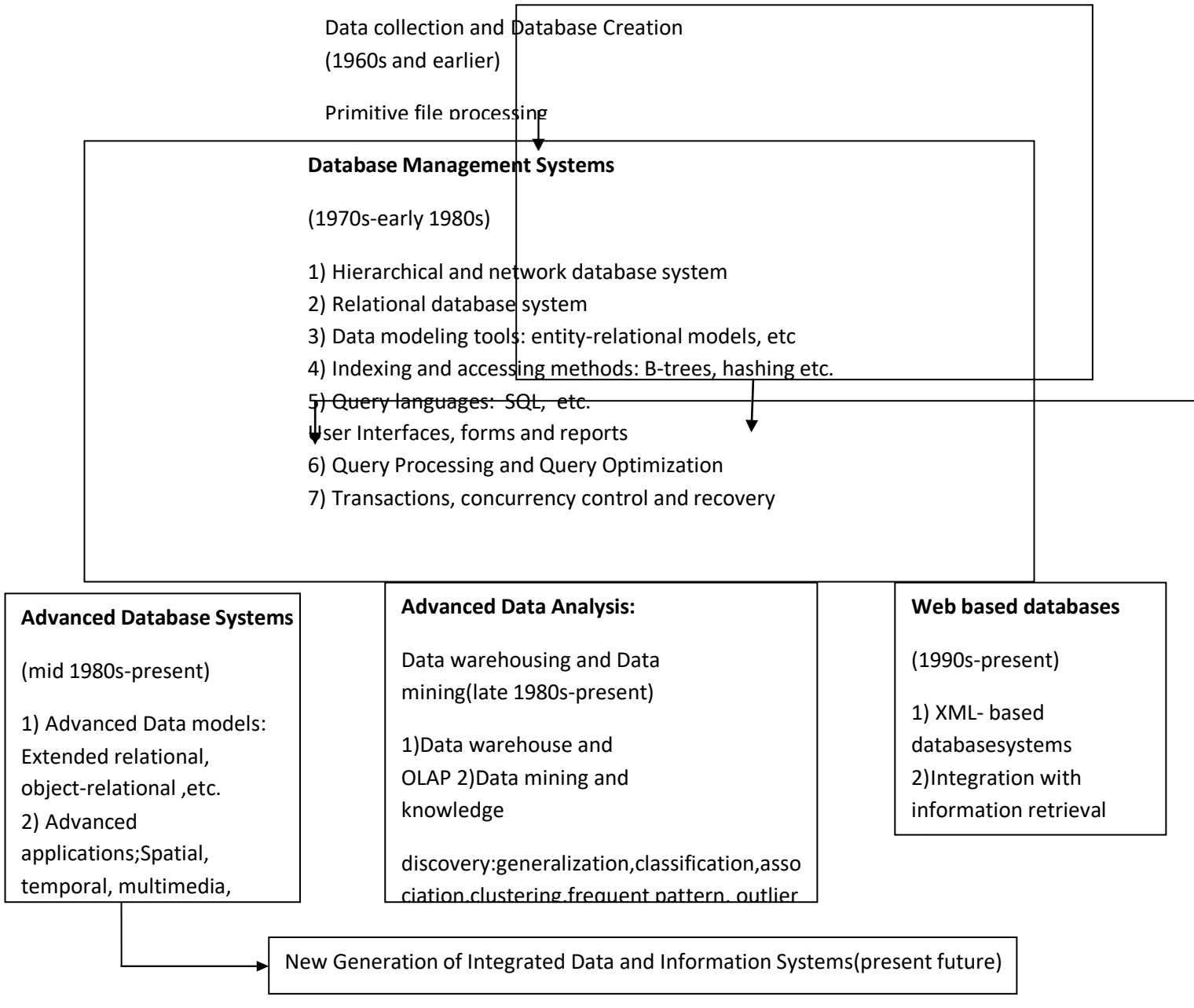
DATA MINING AND WAREHOUSING

LECTURE 1

What motivated data mining? Why is it important?

The major reason that data mining has attracted a great deal of attention in information industry in recent years is due to the wide availability of huge amounts of data and the imminent need for turning such data into useful information and knowledge. The information and knowledge gained can be used for applications ranging from business management, production control, and market analysis, to engineering design and science exploration.

The evolution of database technology



What is data mining?

Data mining refers to extracting or mining" knowledge from large amounts of data. There are many other terms related to data mining, such as knowledge mining, knowledge extraction, data/pattern analysis, data archaeology, and data dredging. Many people treat datamining as a synonym for another popularly used term, Knowledge Discovery in Databases", or KDD

Data mining is the process of seaching and analyzing a large datasets in order to identify patterns and extract useful information.

Essential step in the process of knowledge discovery in databases

/ Steps in data mining process:-

Knowledge discovery as a process is depicted in following figure and consists of aniterative sequence of the following steps:

- data cleaning: to remove noise or irrelevant data
- data integration: where multiple data sources may be combined
- data selection: where data relevant to the analysis task are retrieved from thedatabase
- data transformation: where data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations
- data mining :an essential process where intelligent methods are applied in order toextract data patterns
- pattern evaluation to identify the truly interesting patterns representing knowledge basedon some interestingness measures
- knowledge presentation: where visualization and knowledge representation techniques are used to present the mined knowledge to the user.

Architecture of a typical data mining system/Major Components

Data mining is the process of discovering interesting knowledge from large amounts of data stored either in databases, data warehouses, or other information repositories. Based on this view, the architecture of a typical data mining system may have the following major components:

1. A database, data warehouse, or other information repository, which consists of the set of databases, data warehouses, spreadsheets, or other kinds of information repositories containing the student and course information.

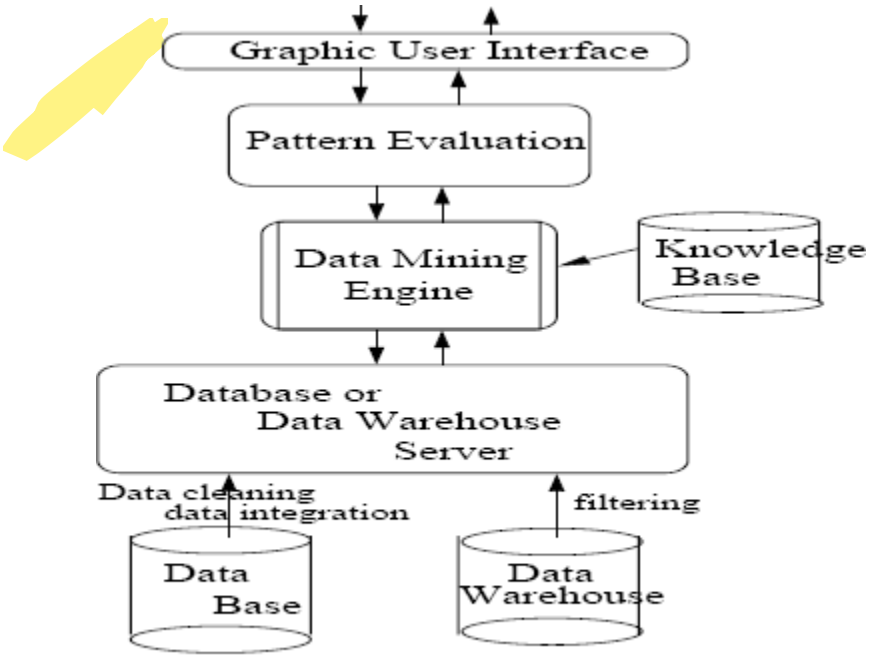
2. A database or data warehouse server which fetches the relevant data based on users' data mining requests.

3. A knowledge base that contains the domain knowledge used to guide the search or to evaluate the interestingness of resulting patterns. For example, the knowledge base may contain metadata which describes data from multiple heterogeneous sources.

4. A data mining engine, which consists of a set of functional modules for tasks such as classification, association, classification, cluster analysis, and evolution and deviation analysis.

5. A pattern evaluation module that works in tandem with the data mining modules by employing interestingness measures to help focus the search towards interestingness patterns.

6. A graphical user interface that allows the user an interactive approach to thedata mining system.



Architecture of a typical data mining system.

How is a data warehouse different from a database? How are they similar?

• Differences between a data warehouse and a database: A data warehouse is a repository of information collected from multiple sources, over a history of time, stored under a unified schema, and used for data analysis and decision support; whereas a database, is a collection of interrelated data that represents the current status of the

stored data. There could be multiple heterogeneous databases where the schema of one database may not agree with the schema of another. A database system supports ad-hoc query and on-line transaction processing. **For more details, please refer to the section "Differences between operational database systems and data warehouses."**

- Similarities between a data warehouse and a database: Both are repositories of information, storing huge amounts of persistent data.

Data mining: on what kind of data? / Describe the following advanced database systems and applications: object-relational databases, spatial databases, text databases, multimedia databases, the World Wide Web.

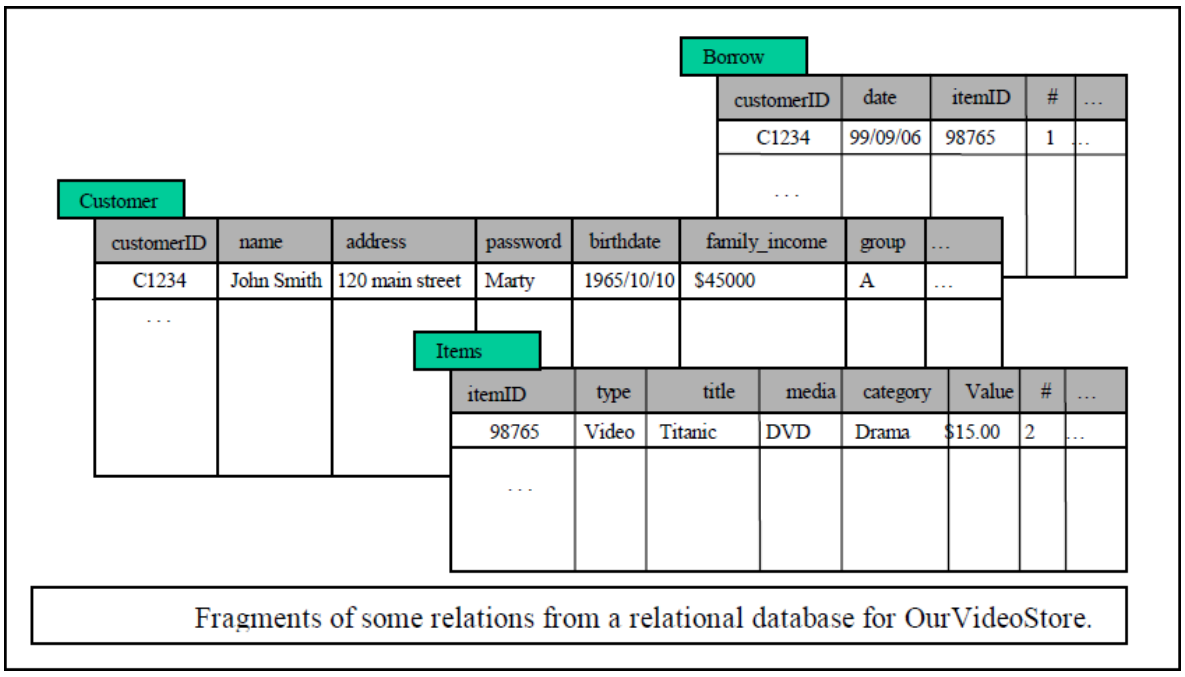
In principle, data mining should be applicable to any kind of information repository. This includes relational databases, data warehouses, transactional databases, advanced database systems,

flat files, and the World-Wide Web. Advanced database systems include object-oriented and object-relational databases, and special c application-oriented databases, such as

spatial databases, time-series databases, text databases, and multimedia databases.

Flat files: Flat files are actually the most common data source for data mining algorithms, especially at the research level. Flat files are simple data files in text or binary format with a structure known by the data mining algorithm to be applied. The data in these files can be transactions, time-series data, scientific measurements, etc.

Relational Databases: a relational database consists of a set of tables containing either values of entity attributes, or values of attributes from entity relationships. Tables have columns and rows, where columns represent attributes and rows represent tuples. A tuple in a relational table corresponds to either an object or a relationship between objects and is identified by a set of attribute values representing a unique key. In following figure it presents some relations Customer, Items, and Borrow representing business activity in a video store. These relations are just a subset of what could be a database for the video storeand is given as an example.



The most commonly used query language for relational database is SQL, which allows retrieval and manipulation of the data stored in the tables, as well as the calculation of aggregate functions such as average, sum, min, max and count.

Features of Data Warehouses

Here is a comprehensive listing of the most common features of data warehouses:

1. **Unified**
   Numerous heterogeneous sources are compiled together to formulate a data warehouse like social databases, level documents, etc.
2. **SubjectOriented**
   It offers the most important data regarding particular subjects like products, suppliers, customers, promotions, etc. It handles data analysis and modelling while assisting organisations in making data-driven decisions
3. **Nonvolatile**
   The previous data remains intact whenever new data is added to the data warehouse. Since the operational database and data warehouse are kept separately, changes within the operational database aren't shown.
4. **Time-Variant**
   Data collected within the data warehouse is seamlessly identified within a particular period.

Features of Data Mining

Some of the unique features of data mining are:

- It is capable of predicting future results
- It can efficiently handle large datasets and databases
- It can seamlessly utilise the automated discovery of patterns
- It has the potential to create actionable insights, etc.

Applications of Data Mining

The applications of data mining include:

- Research
- Education Sector
- Transportation
- Market Basket Analysis
- Business Transactions
- Intrusion Detection
- Scientific Analysis
- Finance and Banking Sector
- Insurance and Healthcare

Applications of Data Warehouses:

The applications of data warehouses include:

- Banking Services
- Consumer Goods
- Manufacturing
- Financial Services
- Retail Sectors

Benefits of Data Mining

Here, we have listed a few potential benefits of data mining:

- Analysing trends within the existing marketplace.
- Detecting typical frauds in phone calls, insurance claims, debit or credit purchases, etc.
- It can make easy predictions within the market before making business decisions.

Benefits of Data Warehousing

The most compelling advantages of data warehousing are:

- Improved performance and productivity
- Cost-effective
- Consistent and accurate data access

Difference Between Data Mining and Data Warehousing

Now, we have understood what is data mining and data warehousing in detail. Here we have listed down the difference between data mining and data warehousing:

Data Mining	Data Warehousing
This procedure involves analysing data patterns	It is exquisitely designed for analytical analysis
Regular data analysis	Periodical data storage
Uses pattern recognition logic to identify patterns	Extracts and stores data to enable easy reporting

It is carried out by business users with the help of engineers	It is carried out by engineers
It helps in extracting data from large data sets	It pools all the relevant data together
Statistics, AI, Machine Learning, and Databases are used in data mining technologies	Integrated, subject-oriented, non-volatile, and time-varying constitute data warehouses
Pattern recognition logic is used for determining patterns	It involves extracting and storing data in perfect order to make efficient reporting
Employs pattern recognition tools to help identify the access patterns	Extracts are stores data in an orderly format, thereby making reporting faster and easier
It helps in creating suggestive patterns of key parameters	When connected with operational business systems like CRM, it deliberately adds value to it

Common Tools and Software Used in Data Warehousing and Data Mining

Let's look at the common tools and software used in data warehousing and data mining: Some of the popular data warehouse tools are:

- Amazon Redshift
- Microsoft Azure
- Google BigQuery
- Snowflake
- Micro Focus Vertica
- Amazon DynamoDB

Some of the popular data mining tools are:

- RapidMiner
- MonkeyLearn
- IBM SPSS Modeler
- Oracle Data Mining
- Knime
- Weka
- Orange
- H2O
- Apache Mahout
- SAS Enterprise Miner

Common Data Mining and Data Warehousing Techniques

Let's look at the common techniques used in data warehousing vs data mining:

The most common techniques of data mining are:

- Association
- Clustering
- Data Visualisation

- Data Cleaning
- Machine Learning
- Classification
- Neural Networks
- Prediction
- Data Warehousing
- Outlier Detection

Common Data Warehousing Techniques
Some of the most common data warehousing techniques are:

- Database Compression
- Columnar Data Storage
- In-Memory Processing
- Massive Parallel Processing (MPP)

LECTURE 2

What Is Data Mining?

Data mining refers to extracting or mining knowledge from large amountsof data. The term is actually a misnomer. Thus, data miningshould have been more appropriately named as knowledge mining which emphasis on mining from large amounts of data.

It is the computational process of discovering patterns in large data sets involving methods at the intersection of artificial intelligence, machine learning, statistics, and database systems.

The overall goal of the data mining process is to extract information from a data set and transform it into an understandable structure for further use.

The key properties of data mining are

- Automatic discovery ofpatterns
- Prediction of likely outcomes
- Creation of actionable information
- Focus on large datasets and databases

The Scope of Data Mining

Data mining derives its name from the similarities between searching for valuable business information in a large database — for example, finding linked products in gigabytes of store scanner data — and mining a mountain for a vein of valuable ore. Both processes require either sifting through an immense amount of material, or intelligently probing it to find exactly where the value resides. Given databases of sufficient size and quality, data mining technology can generate new business opportunities by providing these capabilities:

Automated prediction of trends and behaviors. Data mining automates the process of finding predictive information in large databases. Questions that traditionally required extensive hands-on analysis can now be answered directly from the data — quickly. A typical example of a predictive problem is targeted marketing. Data mining uses data on past promotional mailings to identify the targets most likely to maximize return on investment in future mailings. Other predictive problems include forecasting bankruptcy and other forms of default, and identifying segments of a population likely to respond similarly to given events.

Automated discovery of previously unknown patterns. Data mining tools sweep through databases and identify previously hidden patterns in one step. An example of pattern discovery is the analysis of retail sales data to identify seemingly unrelated products that are often purchased together. Other pattern discovery problems include detecting fraudulent credit card transactions and identifying anomalous data that could represent data entry keying errors.

Tasks of Data Mining

Data mining involves six common classes of tasks:

- **Anomaly detection (Outlier/change/deviation detection)** – The identification of unusual data records, that might be interesting or data errors that require further investigation.

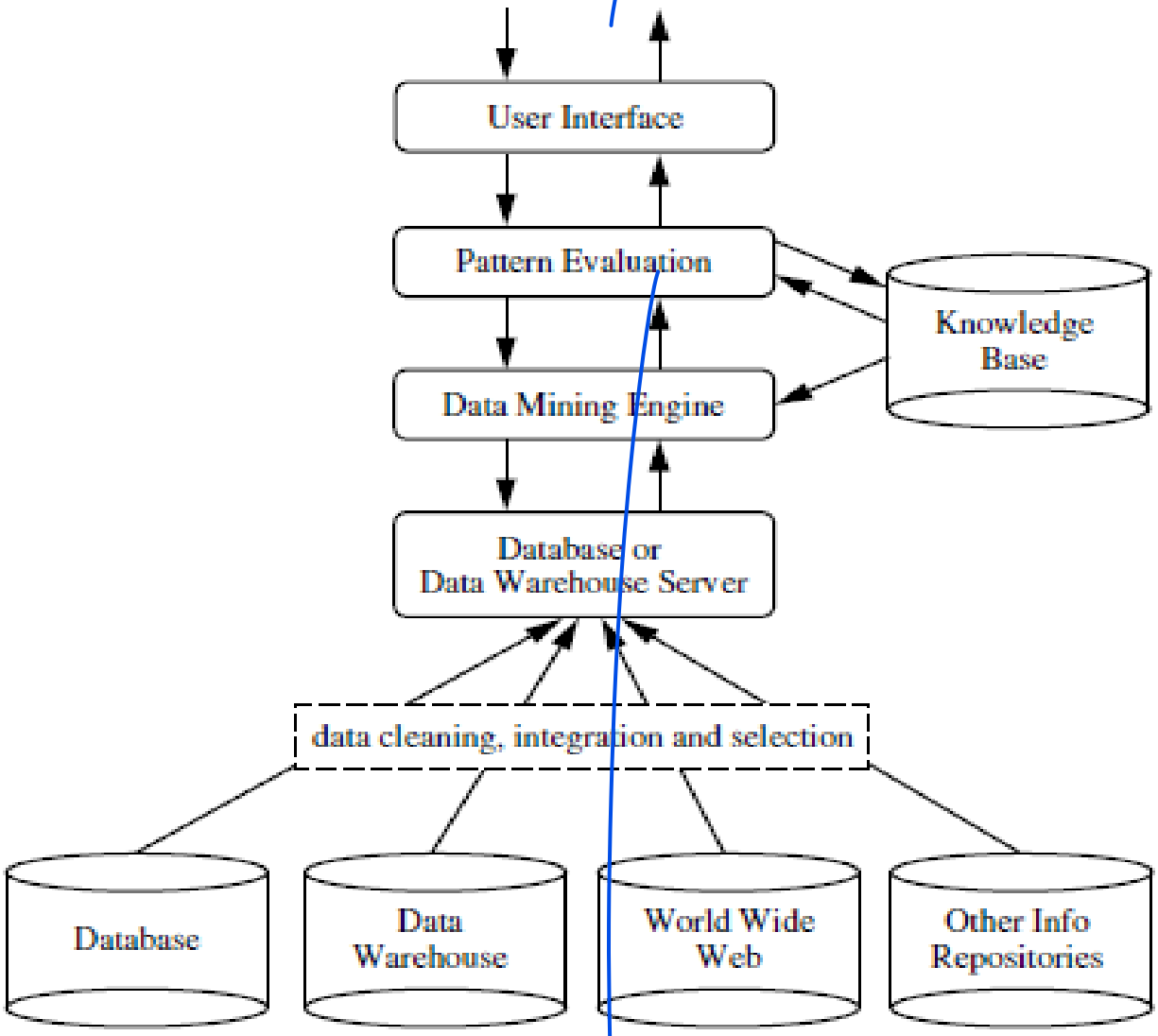
- **Association rule learning (Dependency modelling)** – Searches for relationships between variables. For example a supermarket might gather data on customer purchasing habits. Using association rule learning, the supermarket can determine which products are frequently bought together and use this information for marketing purposes. This is sometimes referred to as market basket analysis.
- **Clustering** – is the task of discovering groups and structures in the data that are in some way or another "similar", without using known structures in the data.

- **Classification** – is the task of generalizing known structure to apply to new data. For example, an e-mail program might attempt to classify an e-mail as "legitimate" or as "spam".
- **Regression** – attempts to find a function which models the data with the least error.

Summarization – providing a more compact representation of the data set, including visualization and report generation.

Architecture of Data Mining

A typical data mining system may have the following major component



1. *Knowledge Base:*

This is the domain knowledge that is used to guide the search orevaluate the interestingness of resulting patterns. Such knowledge can include concepthierarchies, used to organize attributes or attribute values into different levels of abstraction. Knowledge such as user beliefs, which can be used to assess a pattern's interestingness based on its unexpectedness, may also be included. Other examples of domain knowledge are additional interestingness constraints or thresholds, and metadata (e.g., describing data from multiple heterogeneous sources).

1. *Data Mining Engine:*

   This is essential to the data mining systemand ideally consists ofa set of functional modules for tasks such as characterization, association and correlationanalysis, classification, prediction, cluster analysis, outlier analysis, and evolutionanalysis.

2. *Pattern Evaluation Module:*

   This component typically employs interestingness measures interacts with the data mining modules so as to focus thesearch toward interesting patterns. It may use interestingness thresholds to filterout discovered patterns. Alternatively, the pattern evaluation module may be integratedwith the mining module, depending on the implementation of the datamining method used. For efficient data mining, it is highly recommended to pushthe evaluation of pattern interestingness as deep as possible into the mining processso as to confine the search to only the interesting patterns.

3. *User interface:*

Thismodule communicates between users and the data mining system,allowing the user to interact with the system by specifying a data mining query ortask, providing information to help focus the search, and performing exploratory datamining based on the intermediate data mining results. In addition, this componentallows the user to browse database and data warehouse schemas or data structures,evaluate mined patterns, and visualize the patterns in different forms.

LECTURE 3

Data Mining Functionalities

Below are some data mining functionalities:

- **Classification**
  Classification is a supervised learning technique used to categorize data into predefined classes or labels. It involves training a model using a labelled dataset, which is then used to predict the class of new, unlabeled data. Common algorithms used for classification include decision trees, support vector machines, and neural networks

- **Regression**
  Regression is another supervised learning technique aimed at predicting numerical values instead of class labels. It uses historical data to establish a relationship between dependent and independent variables. Linear regression, logistic regression, and polynomial regression are popular regression algorithms

- **Clustering**
  Clustering is an unsupervised learning technique that groups similar data points together based on their features. The goal is to identify underlying structures or patterns in the data. Some common clustering algorithms include K-means, hierarchical clustering, and DBSCAN

- **AssociationRuleMining**
  Association rule mining is a technique used to discover relationships between variables in large datasets. It is commonly used in market basket analysis to identify products that are frequently purchased together. Apriori and Eclat algorithms are popular for association rule mining

- **AnomalyDetection**
  Anomaly detection aims to identify unusual or rare instances in a dataset that deviate from the norm. It is used in various applications, such as fraud detection, network security, and fault diagnosis. Common techniques include statistical methods, machine learning, and deep learning

- **FeatureSelection**
  Feature selection is the process of selecting the most relevant features or attributes from the dataset to improve model performance, reduce complexity, and enhance interpretability. Techniques include filter methods, wrapper methods, and embedded methods

- **DimensionalityReduction**
  Dimensionality reduction reduces the number of features in a dataset while preserving its essential structure and relationships. This can lead to improved model performance, reduced computational complexity, and better visualization. Principal component analysis (PCA) and t-distributed stochastic neighbour embedding (t-SNE) are popular dimensionality reduction techniques

LECTURE 4

Data Mining Process:

Data Mining is a process of discovering various models, summaries, and derived values from a given collection of data.

The general experimental procedure adapted to data-mining problems involves the following steps:

1. State the problem and formulate the hypothesis

Most data-based modeling studies are performed in a particular application domain. Hence, domain-specific knowledge and experience are usually necessary in order to come up with a meaningful problem statement. Unfortunately, many application studies tend to focus on the data-mining technique at the expense of a clear problem statement. In this step, a modeler usually specifies a set of variables for the unknown dependency and, if possible, a general form of this dependency as an initial hypothesis. There may be several hypotheses formulated for a single problem at this stage. The first step requires the combined expertise of an application domain and a data-mining model. In practice, it usually means a close interaction between the data-mining expert and the application expert. In successful data-mining applications, this cooperation does not stop in the initial phase; it continues during the entire data-mining process.

2. Collect the data

This step is concerned with how the data are generated and collected. In general, there are two distinct possibilities. The first is when the data-generation process is under the control of an expert (modeler): this approach is known as a designed experiment. The second possibility is when the expert cannot influence the data- generation process: this is known as the observational approach. An observational setting, namely, random data generation, is assumed in most data-mining applications. Typically, the sampling

distribution is completely unknown after data are collected, or it is partially and implicitly given in the data-collection procedure. It is very important, however, to understand how data collection affects its theoretical distribution, since such a priori knowledge can be very useful for modeling and, later, for the final interpretation of results. Also, it is important to make sure that the data used for estimating a model and the data used later for testing and applying a model come from the same, unknown, sampling distribution. If this is not the case, the estimated model cannot be successfully used in a final application of the results.

3. Preprocessing the data

In the observational setting, data are usually "collected" from the existing databses, data warehouses, and data marts. Data preprocessing usually includes at least two common tasks:

1. **Outlier detection (and removal)** – Outliers are unusual data values that are not consistent with most observations. Commonly, outliers result from measurement errors, coding and recording errors, and, sometimes, are natural, abnormal values. Such nonrepresentative samples can seriously affect the model produced later. There are two strategies for dealing with outliers:

a. Detect and eventually remove outliers as a part of the preprocessing phase, or
b. Develop robust modeling methods that are insensitive to outliers.

1. **Scaling, encoding, and selecting features** – Data preprocessing includes several steps such as variable scaling and different types of encoding. For example, one feature with the range [0, 1] and the other with the range [−100, 1000] will not have the same weights in the applied technique; they will also influence the final data-mining results differently. Therefore, it is recommended to scale them and bring both features to the same weight for further analysis. Also, application-specific encoding methods usually achieve

dimensionality reduction by providing a smaller number of informative features for subsequent data modeling.

These two classes of preprocessing tasks are only illustrative examples of a large spectrum of preprocessing activities in a data-mining process.

Data-preprocessing steps should not be considered completely independent from other data-mining phases. In every iteration of the data-mining process, all activities, together, could define new and improved data sets for subsequent iterations. Generally, a good preprocessing method provides an optimal representation for a data-mining technique by incorporating a priori knowledge in the form of application-specific scaling and encoding.

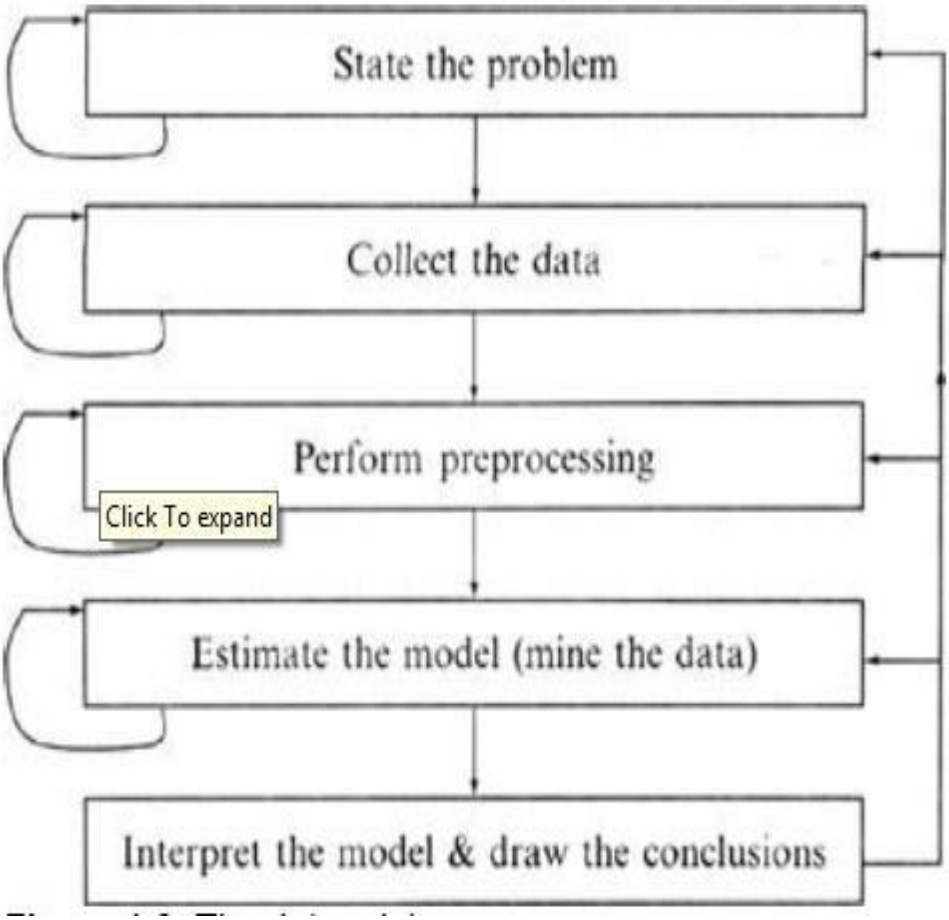
4. *Estimate the model*

The selection and implementation of the appropriate data-mining technique is the main task in this phase. This process is not straightforward; usually, in practice, the implementation is based on several models, and selecting the best one is an additional task. The basic principles of learning and discovery from data are given in Chapter 4 of this book. Later, Chapter 5 through 13 explain and analyze specific techniques that are applied to perform a successful learning process from data and to develop an appropriate model.

5. *Interpret the model and draw conclusions*

In most cases, data-mining models should help in decision making. Hence, such models need to be interpretable in order to be useful because humans are not likely to base their decisions on complex "black-box" models. Note that the goals of accuracy of the model and accuracy of its interpretation are somewhat contradictory. Usually, simple models are more interpretable, but they are also less accurate. Modern data-mining methods are expected to yield highly accurate results using highdimensional models. The problem of interpreting these models, also very important, is considered a separate task, with specific

techniques to validate the results. A user does not want hundreds of pages of numeric results. He does not understand them; he cannot summarize, interpret, and use them for successful decision making.



LECTURE 5

Classification of Data mining Systems:

The data mining system can be classified according to the following criteria:

- Database Technology
- Statistics
- Machine Learning
- InformationScience
- Visualization
- Other Disciplines

Some Other Classification Criteria:

- Classification according to kind of databases mined
- Classification according to kind of knowledge mined
- Classification according to kinds of techniques utilized
- Classification according to applications adapted

Classification according to kind of databases mined

We can classify the data mining system according to kind of databases mined. Database system can be classified according to different criteria such as data models, types of data etc. And the data mining system can be classified accordingly. For example if we classify the database according to data model then we may have a relational, transactional, object- relational, or data warehouse mining system.

Classification according to kind of knowledge mined

We can classify the data mining system according to kind of knowledge mined. It is means data mining system are classified on the basis of functionalities such as:

- Characterization
- Discrimination
- Association and Correlation Analysis
- Classification
- Prediction
- Clustering
- Outlier Analysis
- Evolution Analysis

Classification according to kinds of techniques utilized

We can classify the data mining system according to kind of techniques used. We can describes these techniques according to degree of user interaction involved or the methods of analysis employed.

Classification according to applications adapted

We can classify the data mining system according to application adapted. These applications are as follows:

- Finance
- Telecommunications
- DNA
- Stock Markets
- E-mail

LECTURE 6

Major Issues In Data Mining:

- **Mining different kinds of knowledge in databases.** - The need of different users is not the same. And Different user may be in interested in different kind of knowledge. Therefore it is necessary for data mining to cover broad range of knowledge discovery task.

- **Interactive mining of knowledge at multiple levels of abstraction.** - The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on returned results.

- **Incorporation of background knowledge.** - To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple level of abstraction.

- **Data mining query languages and ad hoc data mining.** - Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.

- **Presentation and visualization of data mining results.** - Once the patterns are discovered it needs to be expressed in high level languages, visual representations. This representations should be easily understandable by the users.

- **Handling noisy or incomplete data.** - The data cleaning methods are required that can handle the noise, incomplete objects while mining the data regularities. If data cleaning methods are not there then the accuracy of the discovered patterns will be poor.

- **Pattern evaluation.** - It refers to interestingness of the problem. The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

- **Efficiency and scalability of data mining algorithms.** - In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.

- **Parallel, distributed, and incremental mining algorithms.** - The factors such as huge size of databases, wide distribution of data,and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithm divide the data into partitions which is further processed parallel. Then the results from the partitions is merged. The incremental algorithms, updates databases without having mine the data again from scratch.

LECTURE 7

What Is Data Wrangling?

Data wrangling is the process of removing errors and combining complex data sets to make them more accessible and easier to analyze. Due to the rapid expansion of the amount of data and data sources available today, storing and organizing large quantities of data for analysis is becoming increasingly necessary.

A data wrangling process, also known as a data munging process, consists of reorganizing, transforming and mapping data from one "raw" form into another in order to make it more usable and valuable for a variety of downstream uses including analytics.

Data wrangling can be defined as the process of cleaning, organizing, and transforming raw data into the desired format for analysts to use for prompt decision-making. Also known as data cleaning or data munging, data wrangling enables businesses to tackle more complex data in less time, produce more accurate results, and make better decisions. The exact methods vary from project to project depending upon your data and the goal you are trying to achieve. More and more organizations are increasingly relying on data wrangling tools to make data ready for downstream analytics.

Importance of Data Wrangling

Did you know, data professionals spend almost 80% of their time wrangling the data, leaving a mere 20% for exploration and modeling?

Some may question if the amount of work and time devoted to data wrangling is worth the effort. A simple analogy will help you understand. The foundation of a skyscraper is expensive and time-consuming before the above-ground structure starts. Still, this solid foundation is extremely valuable for the building to stand tall and serve its purpose for decades. Similarly, for data handling, once the code and infrastructure foundation are gathered, it will deliver immediate results (sometimes almost instantly) for as long as the process is relevant. However, skipping necessary data wrangling steps will lead to significant downfalls, missed opportunities, and erroneous models that damage the reputation of analysis within the organization.

Data wrangling software has become such an indispensable part of data processing. The primary importance of using data wrangling tools can be described as:

- **Making raw data usable. Accurately wrangled data guarantees that quality data is entered into the downstream analysis.**
- **Getting all data from various sources into a centralized location so it can be used.**
- **Piecing together raw data according to the required format and understanding the business context of data**
- **Automated data integration tools are used as data wrangling techniques that clean and convert source data into a standard format that can be used repeatedly according to end requirements. Businesses use this standardized data to perform crucial, cross-data set analytics.**
- **Cleansing the data from the noise or flawed, missing elements**
- **Data wrangling acts as a preparation stage for the data mining process, which involves gathering data and making sense of it.**
- **Helping business users make concrete, timely decisions**

Data wrangling software typically performs six iterative steps of Discovering, Structuring, Cleaning, Enriching, Validating, and Publishing data before it is ready for analytics.

Benefits of Data Wrangling

- **Data wrangling helps to improve data usability as it converts data into a compatible format for the end system.**
- **It helps to quickly build data flows within an intuitive user interface and easily schedule and automate the data-flow process.**
- **Integrates various types of information and their sources (like databases, web services, files, etc.)**
- **Help users to process very large volumes of data easily and easily share data-flow techniques.**

Data Wrangling Tools

There are different tools for data wrangling that can be used for gathering, importing, structuring, and cleaning data before it can be fed into analytics and BI apps. You can use automated tools for data wrangling, where the software allows you to validate data mappings

and scrutinize data samples at every step of the transformation process. This helps to quickly detect and correct errors in data mapping. Automated data cleaning becomes necessary in businesses dealing with exceptionally large data sets. For manual data cleaning processes, the data team or data scientist is responsible for wrangling. In smaller setups, however, non-data professionals are responsible for cleaning data before leveraging it.

Some examples of basic data munging tools are:

- **Spreadsheets / Excel Power Query - It is the most basic manual data wrangling tool**
- **OpenRefine - An automated data cleaning tool that requires programming skills**
- **Tabula – It is a tool suited for all data types**
- **Google DataPrep – It is a data service that explores, cleans, and prepares data**
- **Data wrangler – It is a data cleaning and transforming tool**

Data Wrangling Examples

Data wrangling techniques are used for various use-cases. The most commonly used examples of data wrangling are for:

- **Merging several data sources into one data-set for analysis**
- **Identifying gaps or empty cells in data and either filling or removing them**
- **Deleting irrelevant or unnecessary data**
- **Identifying severe outliers in data and either explaining the inconsistencies or deleting them to facilitate analysis**

Businesses also use data wrangling tools to

- **Detect corporate fraud**
- **Support data security**
- **Ensure accurate and recurring data modeling results**
- **Ensure business compliance with industry standards**
- **Perform Customer Behavior Analysis**
- **Reduce time spent on preparing data for analysis**
- **Promptly recognize the business value of your data**

- **Find out data trends**
  - For example, numeric values for the attribute *income* can be generalized to discrete ranges, such as *low, medium*, and *high*. Similarly, categorical attributes, like *street*, can be generalized to higher-level concepts, like *city*.
  - Data can also be reduced by applying many other methods, ranging from wavelet transformation and principle components analysis to discretization techniques, such as binning, histogram analysis, and clustering.

LECTURE 8

Data Preprocessing: It refers to cleaning, transforming and integrating of data in order to make it ready for analysis .The goal of data preprocessing is to improve the quality of the data and to make it more suitable for the specific data mining task.

Data cleaning

Data Integration:

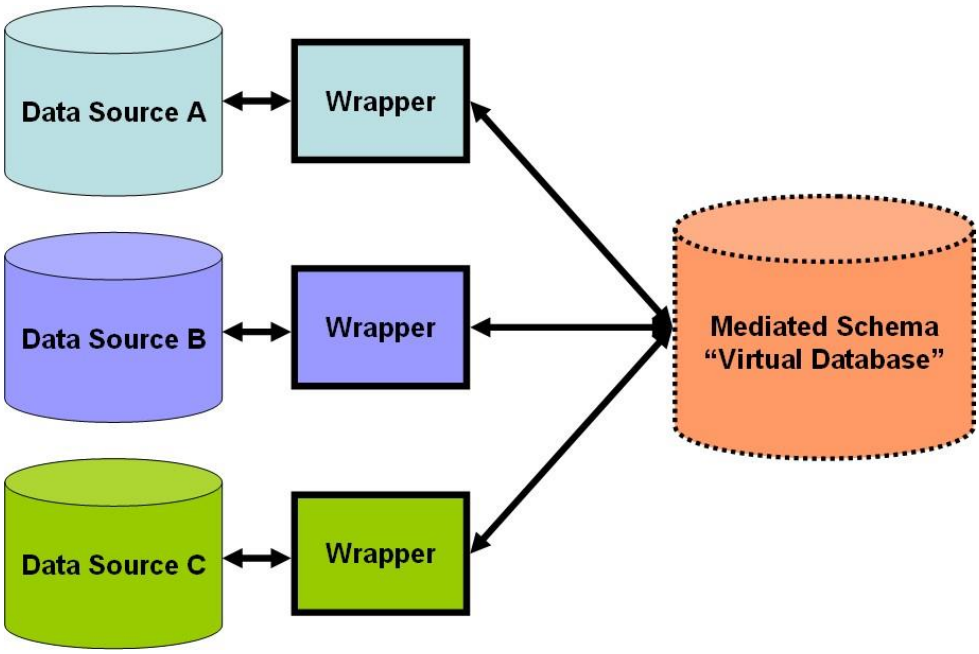
It combines datafrom multiple sources into a coherent data store, as in data warehousing. These sourcesmay include multiple databases, data cubes, or flat files.

The data integration systems are formally defined as triple<G,S,M>

Where G: The global schema

S:Heterogeneous source of schemas

M: Mapping between the queries of source and global schema



Issues in Data integration:

1. *Schema integration and object matching:*

   How can the data analyst or the computer be sure that customer id in one database and customer number in another reference to the same attribute.

2. *Redundancy:*

   An attribute (such as annual revenue, forinstance) may be redundant if it can be derived from another attribute or set ofattributes. Inconsistencies in attribute or dimension naming can also cause redundanciesin the resulting data set.

3. *detection and resolution of datavalue conflicts:*

   For the same real-world entity, attribute values fromdifferent sources may differ.

Data Transformation:

In data transformation, the data are transformed or consolidated into forms appropriatefor mining.

Data transformation can involve the following:

- **Smoothing**, which works to remove noise from the data. Such techniques includebinning,regression, and clustering.
- **Aggregation**, where summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to compute monthly and annualtotal amounts. This step is typically used in constructing a data cube for analysis of the data at multiple granularities.

- **Generalization of the data**, where low-level or ―primitive‖ (raw) data are replaced byhigher-level concepts through the use of concept hierarchies. For example, categoricalattributes, like street, can be generalized to higher-level concepts, like city or country.
- **Normalization**, where the attribute data are scaled so as to fall within a small specifiedrange, such as 1:0 to 1:0, or 0:0 to 1:0.
- **Attribute construction** (or feature construction),wherenewattributes are constructedand added from the given set of attributes to help the mining process.

LECTURE 9

Data Reduction:

Data reduction techniques can be applied to obtain a reduced representation of thedata set that ismuch smaller in volume, yet closely maintains the integrity of the originaldata. That is, mining on the reduced data set should be more efficient yet produce thesame (or almost the same) analytical results.

Strategies for data reduction include the following:

- **Data cube aggregation**, where aggregation operations are applied to the data in theconstruction of a data cube.
- **Attribute subset selection**, where irrelevant, weakly relevant, or redundant attributesor dimensions may be detected and removed.
- **Dimensionality reduction**, where encoding mechanisms are used to reduce the dataset size.
- **Numerosityreduction**,where the data are replaced or estimated by alternative, smallerdata representations such as parametric models (which need store only the modelparameters instead of the actual data) or nonparametric methods such as clustering,sampling, and the use of histograms.
- **Discretization and concept hierarchy generation**,where rawdata values for attributesare replaced by ranges or higher conceptual levels. Data discretization is a form ofnumerosity reduction that is very useful for the automatic generation of concept hierarchies.Discretization and concept hierarchy generation are powerful tools for datamining, in that they allow the mining of data at multiple levels of abstraction.

(i) Data cleaning:

- This refers to the preprocessing of data in order to remove or reduce *noise* (by applying smoothing techniques) and the treatment of *missingvalues* (e.g., by replacing a missing value with the most commonly occurring value for that attribute, or with the most probable value based on statistics).
- Although most classification algorithms have some mechanisms for handling noisy or

missing data, this step can help reduce confusion during learning.

(ii) Data Transformation And Reduction

- The data may be transformed by normalization, particularly when neural networks or methods involving distance measurements are used in the learning step.
- Normalization involves scaling all values for a given attribute so that they fall within a small specified range, such as -1 to +1 or 0 to 1.
- The data can also be transformed by *generalizing* it to higher-level concepts. Concept hierarchies may be used for this purpose. This is particularly useful for continuous valuedattributes.

LECTURE 10

Data Transformation:

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LECTURE 11

Discretization in data mining

Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss. There are two forms of data discretization first is supervised discretization, and the second is unsupervised discretization. Supervised discretization refers to a method in which the class data is used. Unsupervised discretization refers to a method depending upon the way which operation proceeds. It means it works on the top-down splitting strategy and bottom-up merging strategy.

Now, we can understand this concept with the help of an example

Suppose we have an attribute of Age with the given values

Age	1,5,9,4,7,11,14,17,13,18, 19,31,33,36,42,44,46,70,74,78,77

Table before Discretization

Attribute	Age	Age	Age	Age
	1,5,4,9,7	11,14,17,13,18,19	31,33,36,42,44,46	70,74,77,78
After Discretization	Child	Young	Mature	Old

Another example is analytics, where we gather the static data of website visitors. For example, all visitors who visit the site with the IP address of India are shown under country level.

Some Famous techniques of data discretization

Histogram analysis

Histogram refers to a plot used to represent the underlying frequency distribution of a continuous data set. Histogram assists the data inspection for data distribution. For example, Outliers, skewness representation, normal distribution representation, etc.

Binning The original data values are divided in to small intervals known as bins and then they are replaced by a general value calculated for that bin.

Binning refers to a data smoothing technique that helps to group a huge number of continuous values into smaller values. For data discretization and the development of idea hierarchy, this technique can also be used.

Cluster Analysis

Cluster analysis is a form of data discretization. A clustering algorithm is executed by dividing the values of x numbers into clusters to isolate a computational feature of x.

Data discretization using decision tree analysis

Data discretization refers to a decision tree analysis in which a top-down slicing technique is used. It is done through a supervised procedure. In a numeric attribute discretization, first, you need to select the attribute that has the least entropy, and then you need to run it with the help of a recursive process. The recursive process divides it into various discretized disjoint intervals, from top to bottom, using the same splitting criterion.

Data discretization using correlation analysis

Discretizing data by linear regression technique, you can get the best neighboring interval, and then the large intervals are combined to develop a larger overlap to form the final 20 overlapping intervals. It is a supervised procedure.

Data discretization and concept hierarchy generation

The term hierarchy represents an organizational structure or mapping in which items are ranked according to their levels of importance. In other words, we can say that a hierarchy concept refers to a sequence of mappings with a set of more general concepts to complex concepts. It means mapping is done from low-level concepts to high-level concepts. For example, in computer science, there are different types of hierarchical systems. A document is placed in a folder in windows at a specific place in the tree structure is the best example of a computer hierarchical tree model. There are two types of hierarchy: top-down mapping and the second one is bottom-up mapping.

Let's understand this concept hierarchy for the dimension location with the help of an example.

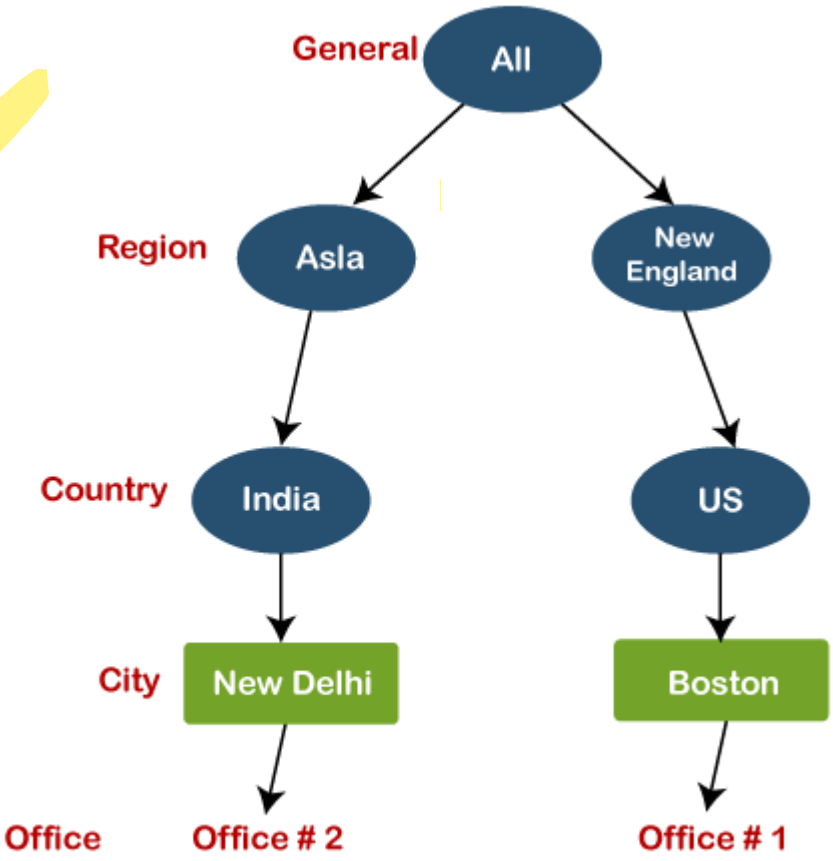
A particular city can map with the belonging country. For example, New Delhi can be mapped to India, and India can be mapped to Asia.

Top-down mapping

Top-down mapping generally starts with the top with some general information and ends with the bottom to the specialized information.

Bottom-up mapping

Bottom-up mapping generally starts with the bottom with some specialized information and ends with the top to the generalized information.



Concept Hierarchy Generation

Data discretization and binarization in data mining

Data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss. In contrast, data binarization is used to transform the continuous and discrete attributes into binary attributes.

Why is Discretization important?

As we know, an infinite of degrees of freedom mathematical problem poses with the continuous data. For many purposes, data scientists need the implementation of discretization. It is also used to improve signal noise ratio.